Go Small or Go Home:

Exploring lightweight generative models for mobile devices

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Introduction

Motivation:

- The growing demand for on-device AI requires efficient generative models for text and image generation on mobile devices.
- Traditional models, while effective, often have high computational and memory costs, making them unsuitable for mobile deployment

Objective:

- We implemented lightweight generative models for both text and images, focusing on optimization techniques such as quantization, pruning, and knowledge distillation.
- The goal is to reduce memory usage, accelerate inference, and maintain output quality, creating highly efficient models suited for mobile hardware constraints

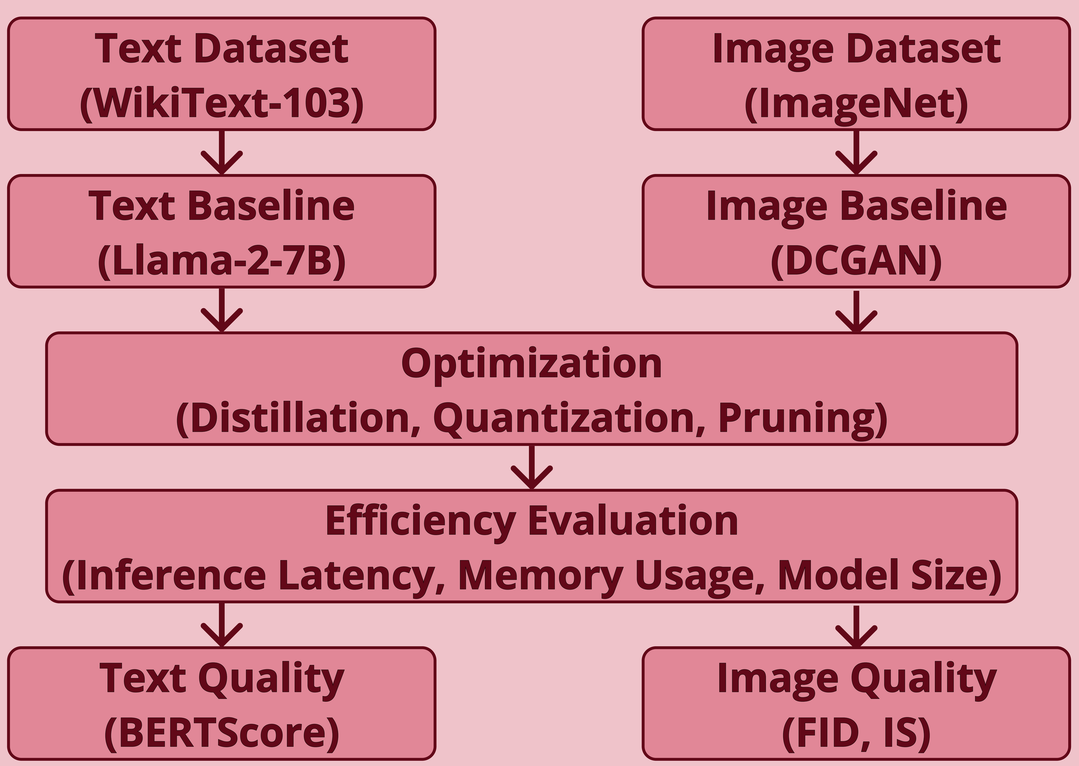
Methodology

Optimization Techniques

Quantization:
Represents model parameters with fewer bits, reducing memory usage and accelerating inference without significant loss in accuracy.

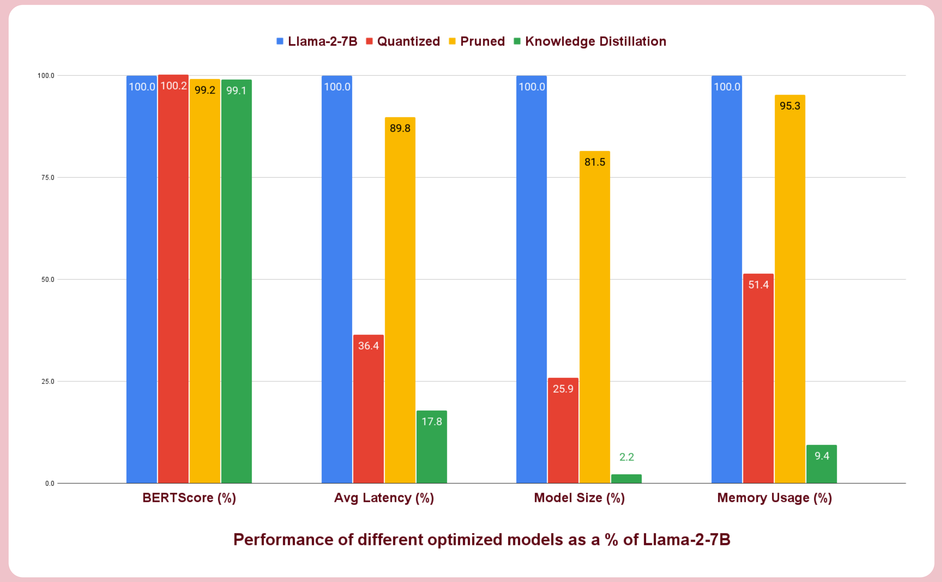
Pruning:
Removes unnecessary weights or connections from neural networks to reduce model size and computational complexity.

Knowledge Distillation:
Transfers knowledge from a larger, complex "teacher" model to a smaller, simpler "student" model, enabling compact models to achieve comparable performance.



Results

Text Generation



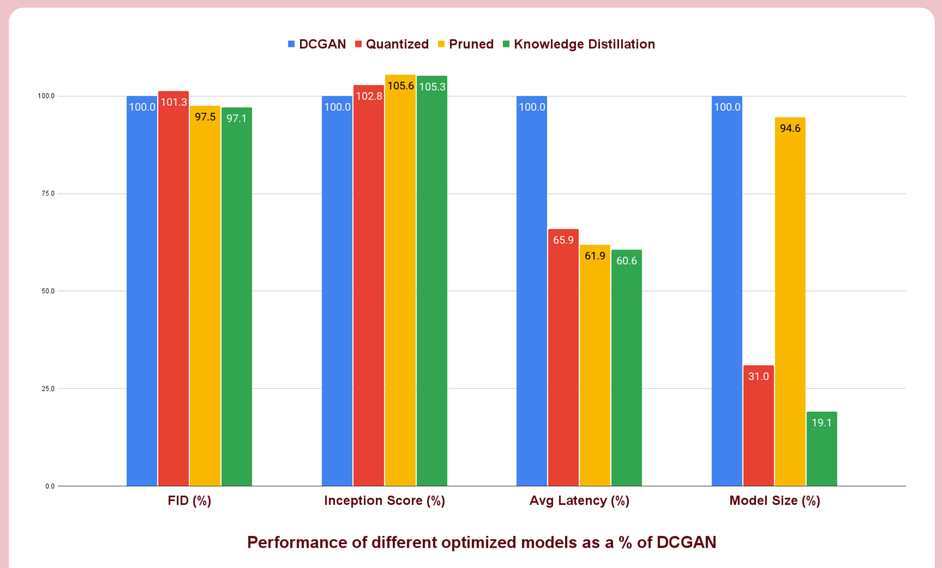
Quality (BERTScore):
All models performed comparable to baseline with a maximum increase of 0.9% in the BERT Score. The quantized model showed improved performance.

Inference Latency:
All models were significantly faster than the baseline. The knowledge distillation was the fastest with an 82% reduction in latency.

Model Size:
All models were significantly smaller than the baseline. The knowledge distillation was the smallest with an 98% reduction in model size.

Memory Usage:
All models were significantly smaller than the baseline. The knowledge distillation was the smallest with an 98% reduction in model size.

Image Generation



Results (contd.)

Quality (Fréchet Inception Distance and Inception Score):
All models performed comparable to or better than the baseline with a maximum increase of 1.3% in the FID while all Inception Scores were better. The pruned model showed improved performance.

Inference Latency:
All models were significantly faster than the baseline. The knowledge distillation was the fastest with a 39% reduction in latency.

Model Size:
All models were significantly smaller than the baseline. The knowledge distillation was the smallest with an 81% reduction in model size.

Conclusion

Minimal Quality Degradation:
All optimization techniques (Knowledge Distillation, Quantization, Pruning) maintain high model quality, with BERTScore remaining above 99% of the baseline and FID/Inception scores showing preserved image generation quality.

Efficiency Improvements:

- **Knowledge Distillation:** 82% faster inference, 98% reduction in model size, 91% decrease in memory usage, with negligible loss in performance.
- **Quantization:** 64% faster inference, 74% reduction in model size, 49% decrease in memory usage, with a very little improvement in performance.
- **Pruning:** 10% faster inference, 19% reduction in model size, 5% decrease in memory usage, with negligible loss in performance.

Conclusion: Mobile-optimized generative AI is feasible through these techniques, offering efficiency gains without sacrificing performance, and allowing for customization based on deployment needs.

References

[1] Angeline Aguinaldo, Ping-Yeh Chiang, Alex Gain, Ameya Patil, Kolten Pearson, and Soheil Feizi. 2019.
Compressing gans using knowledge distillation.

[2] Hanting Chen, Yunhe Wang, Han Shu, Changyuan Wen,Chunjing Xu, Boxin Shi, Chao Xu, and Chang Xu. 2020.
Distilling portable generative adversarial networks for image translation.

[3] Song Han, Huizi Mao, and William J. Dally. 2016.
Deep compression: Compressing deep neural networks with pruning, trained quantization and huffman coding.

[4] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015.
Distilling the knowledge in a neural network.

[5] Xiaoqi Jiao, Yichun Yin, Lifeng Shang, Xin Jiang, Xiao Chen, Linlin Li, Fang Wang, and Qun Liu. 2020.
Tinybert: Distilling bert for natural language understanding.

[6] Daiqing Li, Huan Ling, Amlan Kar, David Acuna, Seung Wook Kim, Karsten Kreis, Antonio Torralba, and Sanja Fidler. 2023.
Dreamteacher: Pretraining image backbones with deep generative models.

[7] Minghan Tang, Ajay Divakaran, and Bryan A. Plummer. 2022.
Image into reasoning: Text-supervised causal scene inference.

[8] Bojun Zhou, Tianyu Cheng, Jiahao Zhao, Chunkai Yan, Ling Jiang, Xinsong Zhang, and Juping Gu. 2024.
Enhancing few-shot learning in lightweight models via dual-faceted knowledge distillation.